Data Warehouse Report

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1 Introduction

The **Online Sales Dataset**[1] is a sample e-commerce database. To support richer analytics the data warehouse includes economic indicators such as *Gross Domestic Product* (GDP) [2] and *GDP per capita*[3] provided from the World Bank.
The aim is to provide multidimensional analytical perspectives relating sales performance to geographical context and economic trends, at a good level of quality, consistency and detail.

2 Dataset Analysis and Preparation

2.1 Primary Dataset Overview

The Online Sales Dataset consists of **49,782 records** across **17 columns**, covering the period from *January 2020* to *September 2025*. Key attributes include:

- **Transaction details**: `InvoiceNo`, `InvoiceDate`, `CustomerID`
- **Product information**: `StockCode`, `Description`, `Category`, `UnitPrice`, `Quantity`, `Discount`
- **Geographic and operational data**: `Country`, `SalesChannel`, `ShipmentProvider`, `OrderPriority`, `ReturnStatus`
- **Financial and logistical metadata**: `ShippingCost`, `PaymentMethod`, `WarehouseLocation`

2.2 Economic Indicators Integration

The World Bank economic datasets contains GDP and GDP per capita indicators in current US dollars for 266 countries from 1960 to 2024. The original dataset is structured as:
`"Country Name","Country Code","Indicator Name","Indicator Code","1960","1961",...,"2024"`

[1] https://www.kaggle.com/datasets/yusufdelikkaya/online-sales-dataset
[2] https://data.worldbank.org/indicator/NY.GDP.MKTP.KD
[3] https://data.worldbank.org/indicator/NY.GDP.PCAP.CD

2.3 Data Quality Assessment

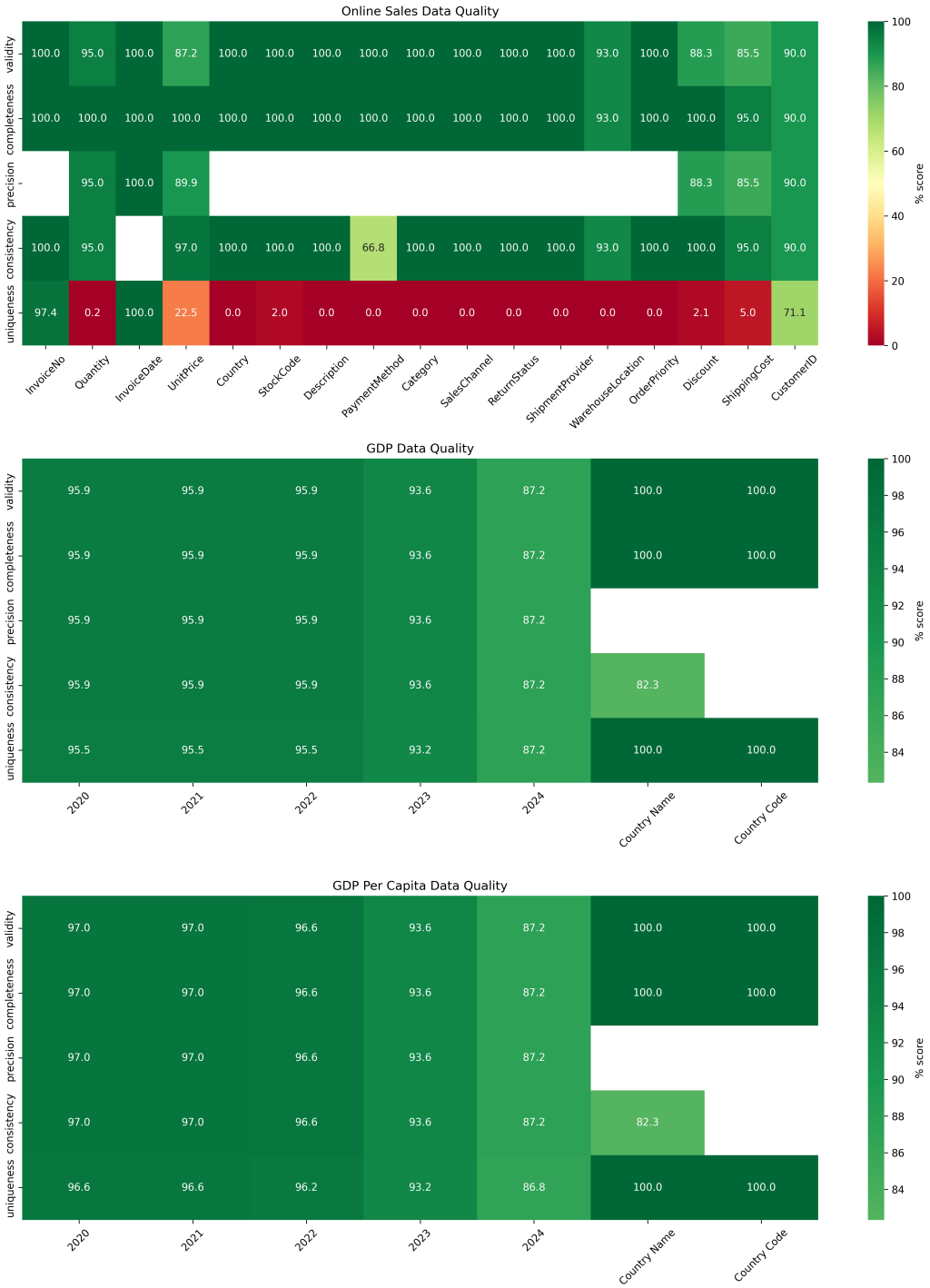

Figure 1: Initial data quality assessment heatmap for the Online Sales Dataset.

Online Sales Dataset Quality Issues:

- **Validity violations**: Fields such as `Quantity` and `UnitPrice` contained invalid negative values; `PaymentMethod` showed typographical errors; `ShipmentProvider`, `Discount`, and `ShippingCost` suffered from inconsistent formats.
- **Completeness issues**: Missing values were present in various fields.

- **Precision anomalies**: Numerical fields exhibited incorrect decimal formatting and inconsistent precision.
- **Consistency Problems**: The synthetic nature of the dataset resulted in random and inconsistent values. Consistency was simulated through approximations and adjustments, which would not be necessary in a real-world dataset.

World Bank Economic Data Quality Issues: The GDP datasets exhibited data gaps primarily in small nation-states and recent years where official statistics had not yet been published.

3 Data Integration with World Bank Economic Indicators

3.1 World Bank Economic Data Processing

The `2_Pulizia_CSV_GDP.py` script was employed to rescale World Bank GDP datasets to a consistent ground. The time period was filtered for values between 2020-2024, matching the range of the Online Sales Dataset.
Data quality improvement involved systematic handling of missing values:

- Linear interpolation was applied where there was minimal missing data to impute missing values
- Those with over two years of consecutive missing data were removed in an attempt to maintain continuity

3.2 Data Cleaning and Preparation Process

The raw datasets required extensive cleaning before integration.

1. **Online Sales Dataset Cleaning**
   Regular expressions were used to filter the correct values. Negative values in numerical fields were deleted or changed in absolute value depending on the type of field. The Discount column was normalized between 0 and 1.
   Text standardization corrected common typos and ensured consistent formatting for InvoiceNo and StockCode values. A significant enhancement was creating realistic multi-item orders. The original dataset contained mostly single-item transactions due to its synthetic nature so we grouped related items into single multi-article orders
2. **Economic Data Integration**
   Integrating the World Bank economic indicators required a careful alignment between the two datasets. The destination countries of the customers in the Online Sales Dataset (OSD) were matched with the countries in the GDP CSV file, using their respective ISO 3166-1 alpha-3 codes. This approach was adopted to save space and simplify representation. Wherever economic data were missing, country-specific interpolation was applied where possible, and after that, the gaps that would still be left were filled by the median of the whole set of data for the world.

Final Dataset Quality The cleaning process transformed the original 49,782 records into a refined dataset of 43,434 records with significantly improved data quality.

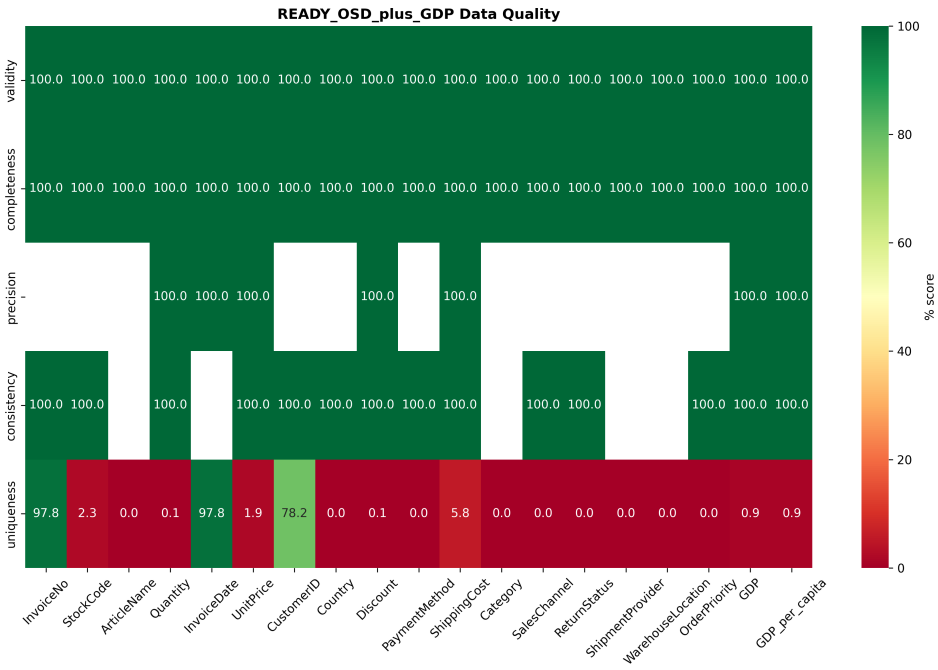

Figure 2: Final data quality assessment heatmap for the cleaned Online Sales Dataset + GDP.

4 Conceptual and Logical Schema

4.1 Initial E/R Schema

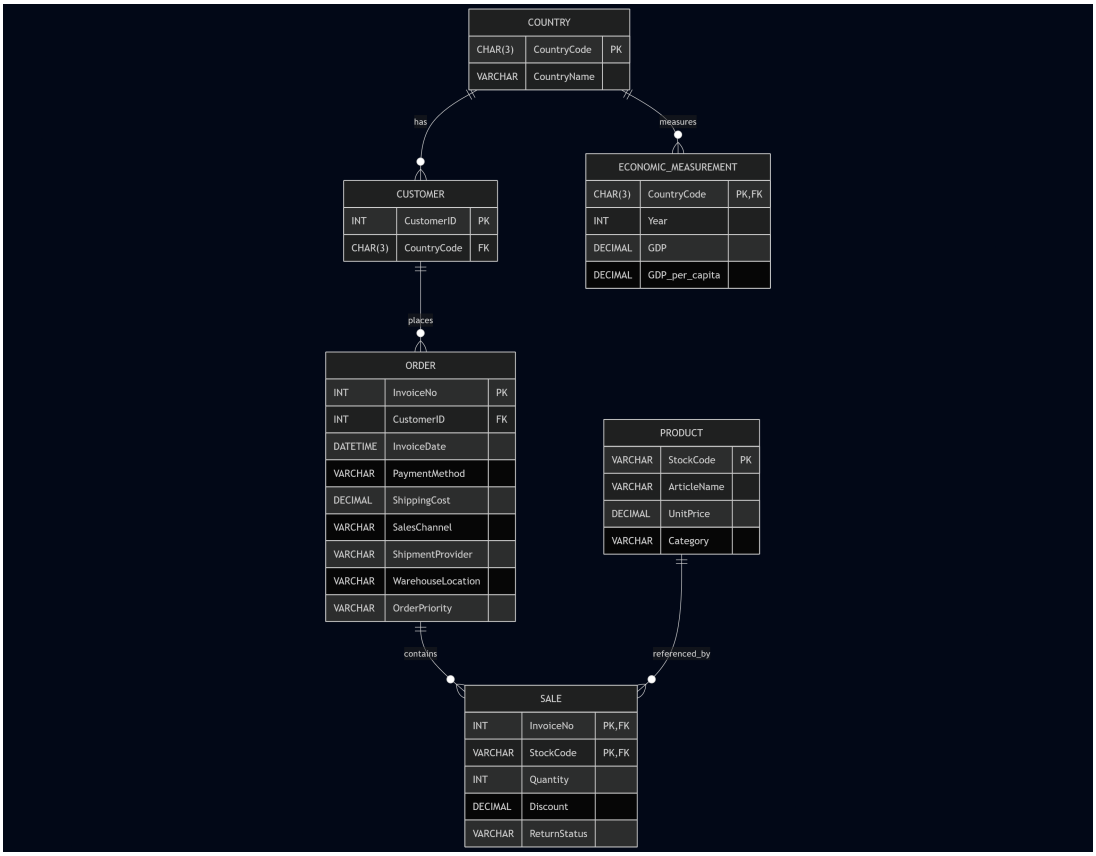

Figure 3: Initial Entity-Relationship diagram of the *Online Sales Dataset*.

The central entities in the first E/R model are: **Sale**, **Order**, **Product**, **Customer**, **Country**, and **Economic_Measurement**. `Order` is central in the schema: each Order is associated with one or more `Sale` entries, each linked to a `Product`. The geographical situation is addressed by the `Customer-Country` relationship, and `Shipment` and `Payment` specify logistics and payment processes, respectively.

4.2 Attribute Tree

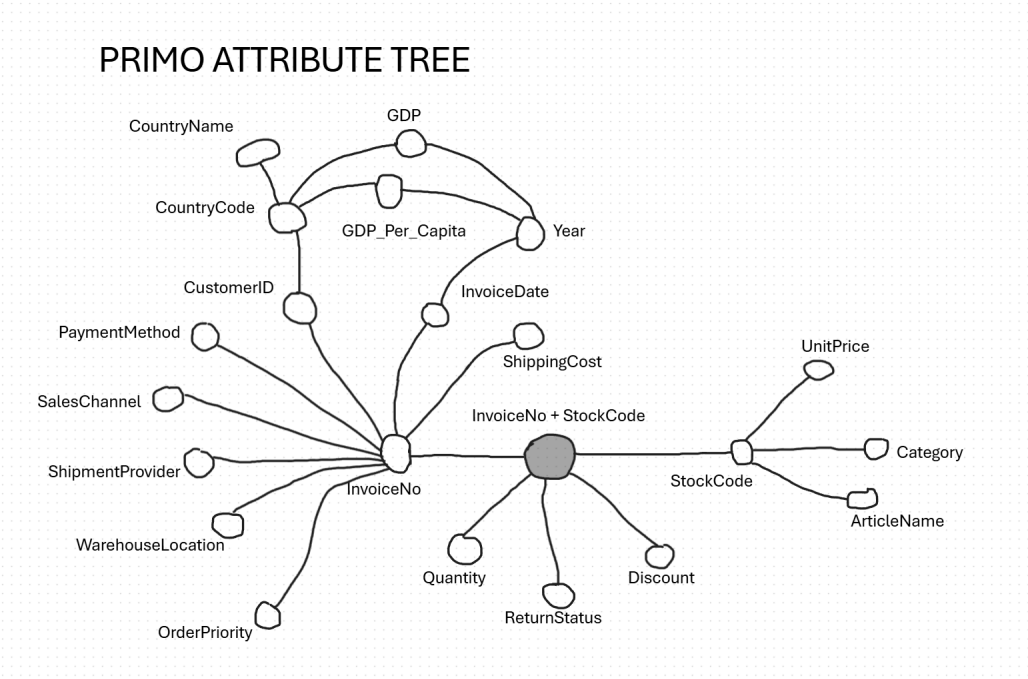

Figure 4: Attribute tree — full version.

Step 1 - initial attribute tree: all the raw dataset attributes were mapped without filters, giving a complete but rough idea of the data available.

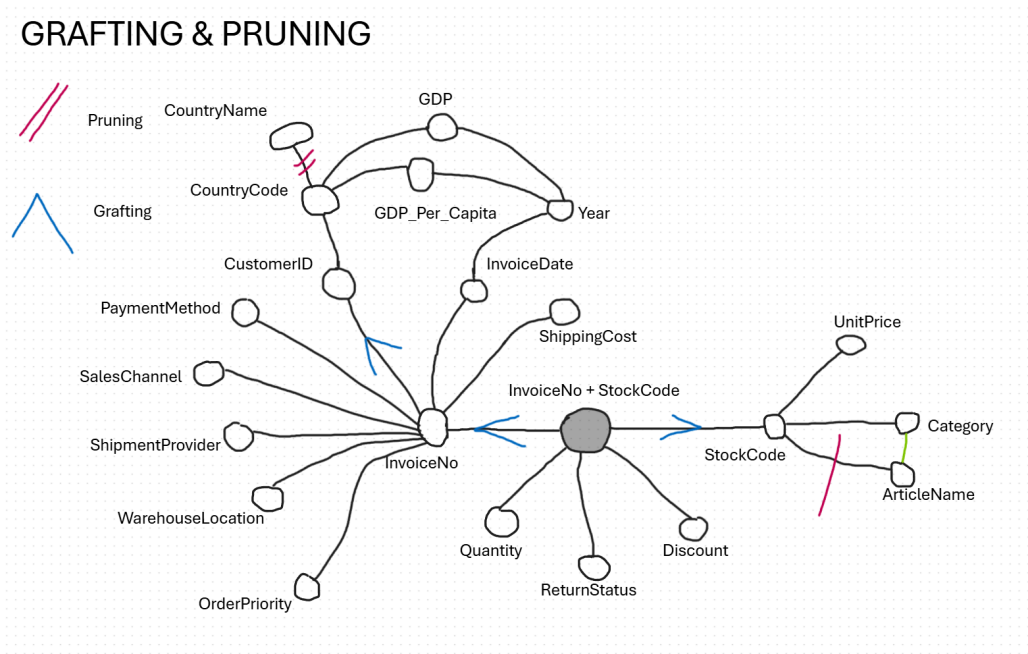

Figure 5: Pruning and grafting of the attribute tree.

Step 2 - pruning and grafting: Attributes such as `InvoiceNo`, `StockCode`, and `CustomerID` were grafted into the tree to enhance its structure. The `CountryName` attribute was pruned as `CountryCode` is sufficient to uniquely identify countries. Additionally, `articleName` was repositioned as part of the hierarchy under `Category`, rather than being directly linked to `StockCode`.

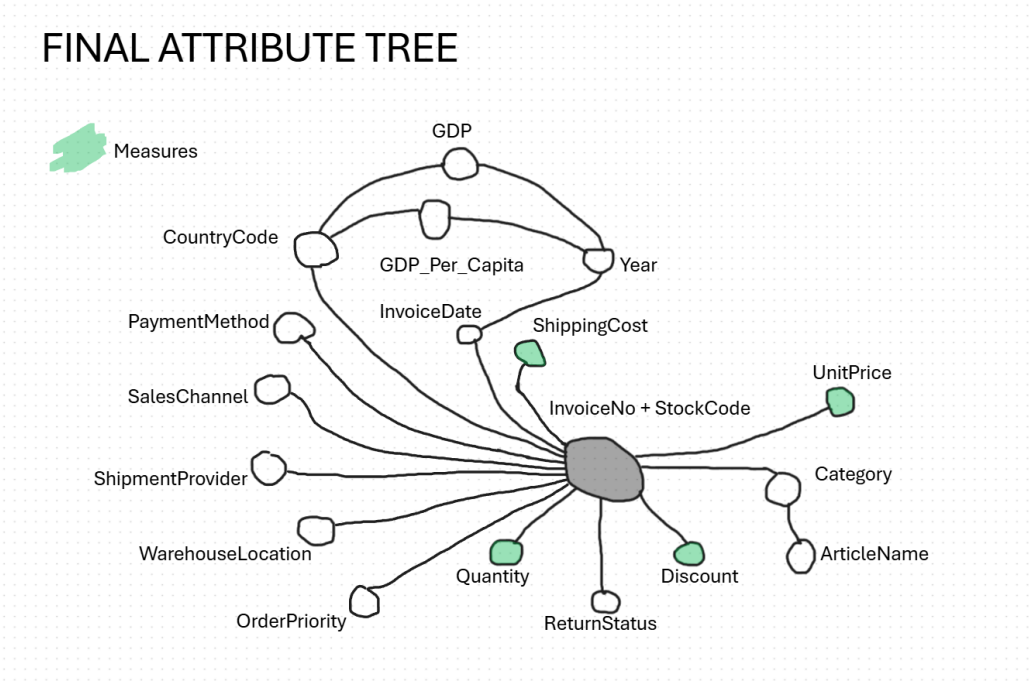

Figure 6: Final attribute tree.

Step 3 - final version: the tree contains only the attributes relevant for analytical purposes, already grouped according to the dimensions planned for the future DW.

4.3 Dimensional Fact Model

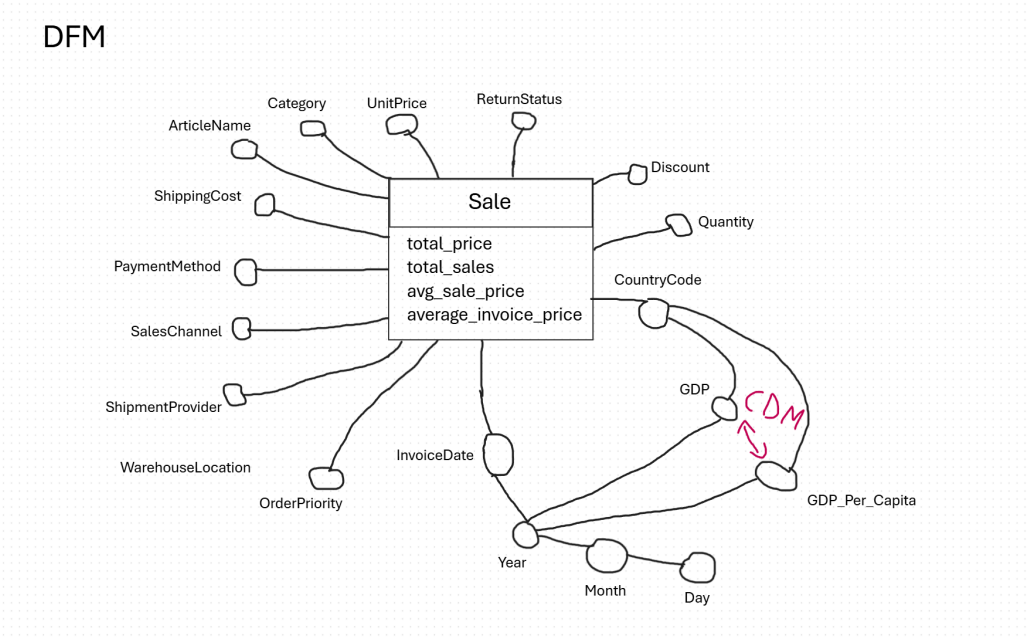

Figure 7: Dimensional Fact Model.

The use of hierarchies, particularly within Product and Date dimensions, facilitates drill-down and roll-up operations. Furthermore, the inclusion of macroeconomic variables as a dimension allows analysis to go beyond transactional data in order to gain perspective on how regional economic trends relate to consumer activity.

4.4 Star Schema

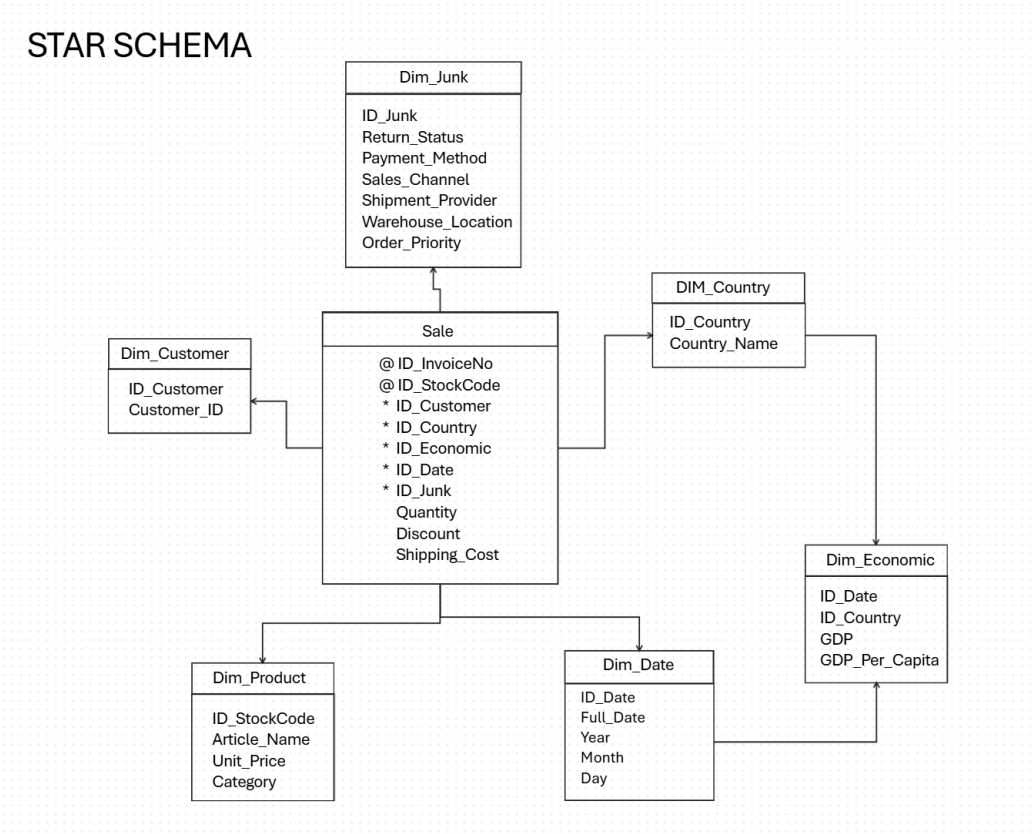

Figure 8: Final Star Schema of the Data Warehouse.

The Star Schema centralizes the analytical model around the `Sale` fact table. The `Dim_Junk` dimension groups low-cardinality attributes (e.g., `Return_Status`, `Sales_Channel`, `Payment_Method`) into a compact structure. The `Dim_Economic` dimension, linked via `ID_Country` and `ID_Date`, enables geo-economic analyses by correlating KPIs with GDP and GDP per capita over time.

References

[1] https://www.kaggle.com/datasets/yusufdelikkaya/online-sales-dataset

[2] https://data.worldbank.org/indicator/NY.GDP.MKTP.KD

[3] https://data.worldbank.org/indicator/NY.GDP.PCAP.CD

[4] https://github.com/PierpaoloSpadafora/Data_Warehouse